

Econometrics I

The multiple linear regression model in Matrix form

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The linear regression model in matrix form

- Goals:
 - ▶ Provide a compact and useful description of the multiple linear regression model using matrices.
 - ▶ Derive matrix expressions for the OLS estimators, its expected values, variances and covariances.
 - ▶ Use the new notation to give a simple proof of the unbiasedness of the estimators.
- Understanding the new notation will be extremely useful for the next topics (deriving the sampling distribution of the OLS estimators and hypothesis testing.)
- Review appendix D from the book for a summary of matrix algebra.

- Consider the multiple linear regression model with k explanatory variables plus a constant:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + u_i, \quad i = 1, \dots, n$$

- In what follows, bold typeface will represent vectors and matrices.
- Define for each observation i the following $(k+1) \times 1$ column vector $\mathbf{x}_i = (1, x_{i1}, x_{i2}, \dots, x_{ik})'$ and let $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_k)'$ be the $(k+1) \times 1$ column vector of the parameters.
- The multiple linear regression model can be written as $y_i = \mathbf{x}_i' \boldsymbol{\beta} + u_i, \quad i = 1, \dots, n$
- Define $\mathbf{y}$ as the $n \times 1$ vector that collects all the observations of the dependent variable, $\mathbf{u}$ for the vector that collects the disturbance or error terms and define the matrix of explanatory variables $\mathbf{X}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1' \\ \mathbf{x}_2' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \vdots & \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

- Then the multiple regression model can be written in *matrix notation* as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

- The least squares estimator is obtained by minimizing the sum of squared residuals, which is the following quadratic form

$$SSR(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b})$$

- The OLS estimator is obtained by solving the *first order condition*¹

$$\frac{\partial SSR(\hat{\boldsymbol{\beta}})}{\partial \mathbf{b}} = -2\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0$$

- Thus we have the following set of *normal equations or first order conditions for OLS*

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$$

¹Remember, here we are using *matrix derivatives*, which is just a short-hand way to collect the set of partial derivatives with respect to each of the components of the vector $\mathbf{b}$. Review the other set of slides to see how this looks like without matrix notation.

- The set of equations in matrix form $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$ is a system of $k + 1$ linear equations with $k + 1$ unknowns.
- Define $\mathbf{Q} = \mathbf{X}'\mathbf{X}$ and rewrite the previous set of equations as $\mathbf{Q}\hat{\beta} = \mathbf{X}'\mathbf{y}$.
- $\mathbf{Q}$ is a $(k + 1) \times (k + 1)$ symmetric matrix of sums of squares and cross-products of explanatory variables, while $\mathbf{X}'\mathbf{y}$ is the $k + 1 \times 1$ vector of sums of cross-products of the explanatory variables and the dependent variable.
- Here are two cases to consider:
 - 1 Full-rank case: when $\mathbf{Q}$ is invertible, i.e., has rank $k + 1$.
 - 2 Short-rank case: when $\mathbf{Q}$ is not invertible, i.e., its rank is less than $k + 1$.
- In the full rank case, the normal equations have a unique solution, the OLS estimator:

$$\hat{\beta} = \mathbf{Q}^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \quad (1)$$

Some important matrices

- Define $\mathbf{A} = \mathbf{Q}^{-1}\mathbf{X}'$, then the *OLS coefficients* are $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}$
- Define $\mathbf{N} = \mathbf{X}\mathbf{A} = \mathbf{X}'\mathbf{Q}^{-1}\mathbf{X}$, then the *predicted values vector* is $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}\mathbf{A}\mathbf{y} = \mathbf{N}\mathbf{y}$. The matrix $\mathbf{N}$ is $n \times n$.
- The *residual vector* is $\hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{I}\mathbf{y} - \mathbf{N}\mathbf{y} = (\mathbf{I} - \mathbf{N})\mathbf{y} = \mathbf{M}\mathbf{y}$, where $\mathbf{M} = \mathbf{I} - \mathbf{N}$ is $n \times n$, and $\mathbf{I}$ is the $n \times n$ identity matrix.
- Note how each of the previous matrices operate on the vector $\mathbf{y}$:
 - ▶ $\mathbf{A}$ transforms the vector of dependent observations into the OLS coefficients.
 - ▶ $\mathbf{N}$ transforms the vector of dependent observations into predicted or fitted values.
 - ▶ $\mathbf{M}$ transforms the vector of dependent observations into residuals.
- The minimized sum of squared residuals is then $SSR(\hat{\boldsymbol{\beta}}) = \hat{\mathbf{u}}'\hat{\mathbf{u}}$

Regression Algebra

- Note that the first order condition implies that $\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$.
- The matrices $\mathbf{Q}$, $\mathbf{A}$, $\mathbf{N}$ and $\mathbf{M}$ have the following properties that will prove useful in the future:

$$(\mathbf{Q}^{-1})' = \mathbf{Q}^{-1} \quad (2)$$

$$\mathbf{AX} = (\mathbf{Q}^{-1}\mathbf{X}')\mathbf{X} = \mathbf{Q}^{-1}\mathbf{Q} = \mathbf{I} \quad (3)$$

$$\mathbf{AA}' = (\mathbf{Q}^{-1}\mathbf{X}')(\mathbf{X}(\mathbf{Q}^{-1})') = \mathbf{Q}^{-1}\mathbf{Q}\mathbf{Q}^{-1} = \mathbf{Q}^{-1} \quad (4)$$

$$\mathbf{N}' = \mathbf{A}'\mathbf{X}' = (\mathbf{X}(\mathbf{Q}^{-1})')\mathbf{X}' = \mathbf{X}\mathbf{Q}^{-1}\mathbf{X}' = \mathbf{N} \quad (5)$$

$$\mathbf{NN} = (\mathbf{XA})\mathbf{XA} = \mathbf{X}(\mathbf{AX})\mathbf{A} = \mathbf{XIA} = \mathbf{XA} = \mathbf{N} \quad (6)$$

$$\mathbf{M}' = \mathbf{I} - \mathbf{N}' = \mathbf{I} - \mathbf{N} = \mathbf{M} \quad (7)$$

$$\mathbf{MM} = (\mathbf{I} - \mathbf{N})(\mathbf{I} - \mathbf{N}) = \mathbf{I} - \mathbf{N} - \mathbf{N} + \mathbf{NN} = \mathbf{I} - \mathbf{N} = \mathbf{M} \quad (8)$$

$$\mathbf{NX} = (\mathbf{XA})\mathbf{X} = \mathbf{X}(\mathbf{AX}) = \mathbf{XI} = \mathbf{X} \quad (9)$$

$$\mathbf{MX} = (\mathbf{I} - \mathbf{N})\mathbf{X} = \mathbf{X} - \mathbf{NX} = \mathbf{X} - \mathbf{X} = \mathbf{O} \quad (10)$$

- Above, the matrix $\mathbf{O}$ is a matrix (the appropriate size should be understood from the context) full of zeros.
- Observe that the matrix $\mathbf{Q}^{-1}$ is symmetric and the matrices $\mathbf{N}$ and $\mathbf{M}$ are idempotent.

Regression algebra

- Properties 9 and 10 deserve a little bit more attention, let us look at what they are telling us.
- Recall that the $\mathbf{N}$ and $\mathbf{M}$ matrices transform the vector of observations of the dependent variable $\mathbf{y}$ into the fitted values vector and the residuals vector respectively.
- This property is true for any vector: by this I mean that if I feed an arbitrary vector $\mathbf{z}$ to $\mathbf{N}$ and $\mathbf{M}$, what I will get is:
 - ▶ From $\mathbf{Nz}$: a vector $\hat{\mathbf{z}}$ of fitted values, which can be written as a *linear combination of the columns of $\mathbf{X}$* and in which the weights for this columns are the estimated coefficients given by $\mathbf{b}_z = \mathbf{AZ}$.
 - ▶ From $\mathbf{Mz}$: a vector of residuals $\hat{\mathbf{u}}_z$ which records the deviations from the true value of each of the components of $\mathbf{z}$ with $\hat{\mathbf{z}}$
- With the previous observations now we can understand properties 9 and 10:
 - ▶ Property 9 is telling you that for any column of the matrix $\mathbf{X}$ of explanatory variables, the best way to write this column as a linear combination of the columns of $\mathbf{X}$ is by **using the column itself**.
 - ▶ Because of the above, then the residual of any column of $\mathbf{X}$ with respect to its fitted value vector is the zero vector.
 - ▶ Properties 9 and 10 collect the previous two facts in matrix form: they are true for any column of the matrix of explanatory variables.

The rank of $\mathbf{X}$ and $\mathbf{Q}$

- When I discussed the first order conditions, I distinguished two cases, the full-rank case and the short-rank case.
- It turns out we have the following result:

Proposition 1

$\text{rank}(\mathbf{Q}) = \text{rank}(\mathbf{X})$, i.e., the rank of the matrix $\mathbf{Q}$ is the same as the rank of the matrix $\mathbf{X}$.

- This means that we can find a unique solution to the first order conditions by inverting the matrix $\mathbf{Q}$ when it has full rank, i.e. $\text{rank}(\mathbf{Q}) = k + 1$, and this rank is equal to $\text{rank}(\mathbf{X})$, that is, the number of linearly independent column of the matrix $\mathbf{X}$.

Matrix Algebra for random variables

- Before going further towards the analysis of the statistical properties of the OLS estimator, we need to introduce a general matrix-algebra system for dealing with random variables.
- Suppose that we have a set $Y_1, \dots, Y_n$ of n random variables whose joint pdf (or pmf) is $f(y_1, \dots, y_n)$.
- The expectations, variances and covariances are, for each $i, h = 1, \dots, n$:

$$E(Y_i) = \mu_i, \quad V(Y_i) = \sigma_i^2 = \sigma_{ii}, \quad C(Y_h, Y_i) = \sigma_{hi} = \sigma_{ih}.$$

- It is natural to display this in a $n \times 1$ vector $\mathbf{Y}$, a $n \times 1$ vector $\boldsymbol{\mu}$, and an $n \times n$ matrix $\boldsymbol{\Sigma}$:

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \sigma_{hi} & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix}$$

- At the risk of some confusion, I will adopt the matrix-algebra convention of lowercase characters for vectors, overriding the statistical convention of uppercase characters for random variables, and write the $n \times 1$ random vector and its elements as $\mathbf{y} = (y_1, \dots, y_n)'$

- The expectation of a random vector (or matrix) is defined to be the vector (or matrix) of expectations of its elements.
- The variance matrix of a random vector is defined to be the matrix of variances and covariances of its elements.
- Write $E(\mathbf{y}) = \boldsymbol{\mu}$ and $V(\mathbf{y}) = \boldsymbol{\Sigma}$.
- When $\mathbf{y}$ has dimension $n \times 1$, $\boldsymbol{\mu}$ has the same dimension and $\boldsymbol{\Sigma}$ is $n \times n$ and symmetric.
- Let $\boldsymbol{\epsilon} = \mathbf{y} - \boldsymbol{\mu}$ be the $n \times 1$ vector of deviations of the y 's from their respective expectations. Thus

$$\boldsymbol{\epsilon}\boldsymbol{\epsilon}' = (\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})' = \{(y_h - \mu_h)(y_i - \mu_i)\}_{i,h=1,\dots,n}$$

is an $n \times n$ symmetric random matrix whose elements are the squares and cross-products of those deviations.

- Then

$$E(\boldsymbol{\epsilon}) = E(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$$

$$E(\boldsymbol{\epsilon}\boldsymbol{\epsilon}') = E[(\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})'] = \boldsymbol{\Sigma} = V(\mathbf{y}) = V(\boldsymbol{\epsilon})$$

- The covariance matrix of a pair of random vectors is defined to be the matrix of covariances between the elements of one vector and the elements of another vector. Thus, if $\mathbf{z} = \{z_i\}_{i=1,\dots,m}$ and $\mathbf{y} = \{y_i\}_{i=1,\dots,n}$ are a pair of random vectors with lengths m and n respectively, the

$$C(\mathbf{z}, \mathbf{y}) = E\{[\mathbf{z} - E(\mathbf{z})][\mathbf{y} - E(\mathbf{y})]'\}$$

is the $m \times n$ matrix whose (h, i) -th entry is $C(z_h, y_i)$, while $C(\mathbf{y}, \mathbf{z})$ is the $n \times m$ transpose of that matrix.

- We will provide some rules for calculating expectations, variances, and covariances of certain functions of $\mathbf{y}$.
- In what follows, $\mathbf{y}$ is $n \times 1$, it has expectation $E(\mathbf{y}) = \boldsymbol{\mu}$, variance matrix $V(\mathbf{y}) = \boldsymbol{\Sigma}$ and write $\boldsymbol{\epsilon} = \mathbf{y} - \boldsymbol{\mu}$.
- Once again, review appendix D for a review of matrix terminology.

Some rules to compute expectations, variances and covariances of random vectors

Proposition 2 (Scalar Linear Function)

Let $z = g + \mathbf{h}'\mathbf{y}$, where the scalar g and the $n \times 1$ vector $\mathbf{h}$ are non-random constants. Then the random variable z has

$$E(z) = g + \mathbf{h}'E(\mathbf{y}) = g + \mathbf{h}'\boldsymbol{\mu}$$

Let $z^* = z - E(z)$, then $z^* = \mathbf{h}'\mathbf{y} - \mathbf{h}'\boldsymbol{\mu} = \mathbf{h}'(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{h}'\boldsymbol{\epsilon}$ and $z^{*2} = (\mathbf{h}'\boldsymbol{\epsilon})^2 = (\mathbf{h}'\boldsymbol{\epsilon})(\mathbf{h}'\boldsymbol{\epsilon}) = (\mathbf{h}'\boldsymbol{\epsilon})(\boldsymbol{\epsilon}'\mathbf{h}) = \mathbf{h}'\boldsymbol{\epsilon}\boldsymbol{\epsilon}'\mathbf{h}$. So

$$V(z) = E(z^{*2}) = E(\mathbf{h}'\boldsymbol{\epsilon}\boldsymbol{\epsilon}'\mathbf{h}) = \mathbf{h}'E(\boldsymbol{\epsilon}\boldsymbol{\epsilon}')\mathbf{h} = \mathbf{h}'\boldsymbol{\Sigma}\mathbf{h}.$$

Since $V(z)$ needs to be nonnegative and $\mathbf{h}$ is an arbitrary $n \times 1$ vector, we see that every variance matrix $\boldsymbol{\Sigma}$ is nonnegative definite, and it is positive definite if and only if it is nonsingular.

Proposition 3 (Vector linear function)

Let $\mathbf{z} = \mathbf{g} + \mathbf{H}\mathbf{y}$ where the $k \times 1$ vector $\mathbf{g}$ and $k \times n$ matrix $\mathbf{H}$ are non-random. Then the $k \times 1$ random vector $\mathbf{z}$ has

$$E(\mathbf{z}) = \mathbf{g} + \mathbf{H}\boldsymbol{\mu}.$$

Further, let $\mathbf{z}^* = \mathbf{z} - E(\mathbf{z})$. Then $\mathbf{z}^* = \mathbf{H}(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{H}\boldsymbol{\epsilon}$, and $\mathbf{z}^*\mathbf{z}^{*'} = \mathbf{H}\boldsymbol{\epsilon}\boldsymbol{\epsilon}'\mathbf{H}'$. So

$$V(\mathbf{z}) = E(\mathbf{z}^*\mathbf{z}^{*'}) = E(\mathbf{H}\boldsymbol{\epsilon}\boldsymbol{\epsilon}'\mathbf{H}') = \mathbf{H}E(\boldsymbol{\epsilon}\boldsymbol{\epsilon}')\mathbf{H}' = \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}'.$$

Proposition 4 (Mean Squares)

Let $\mathbf{W} = \mathbf{y}\mathbf{y}'$. Then the $n \times n$ random matrix $\mathbf{W}$ has expectation $E(\mathbf{W}) = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}'$

Proof.

Write $\mathbf{y}\mathbf{y}' = (\boldsymbol{\mu} + \boldsymbol{\epsilon})(\boldsymbol{\mu} + \boldsymbol{\epsilon})' = \boldsymbol{\mu}\boldsymbol{\mu}' + \boldsymbol{\mu}\boldsymbol{\epsilon}' + \boldsymbol{\epsilon}\boldsymbol{\mu}' + \boldsymbol{\epsilon}\boldsymbol{\epsilon}'$, which, since $\boldsymbol{\mu}$ is constant and $E(\boldsymbol{\epsilon}) = 0$, implies $E(\mathbf{W}) = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}'$ □

Proposition 5 (Sum of squares)

Let $w = \mathbf{y}'\mathbf{y}$. Then the scalar random variable w has expectation $E(w) = \text{trace}(\mathbf{\Sigma}) + \boldsymbol{\mu}'\boldsymbol{\mu}$, where tr is the trace operator.

Proof.

Write $\mathbf{y}'\mathbf{y} = \text{trace}(\mathbf{y}'\mathbf{y}) = \text{trace}(\mathbf{y}\mathbf{y}') = \text{trace}(\mathbf{W})$, so

$$\begin{aligned} E(\mathbf{y}'\mathbf{y}) &= E[\text{trace}(\mathbf{W})] = tr[E(\mathbf{W})] = \text{trace}(\mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}') \\ &= \text{trace}(\mathbf{\Sigma}) + \text{trace}(\boldsymbol{\mu}\boldsymbol{\mu}') = \text{trace}(\mathbf{\Sigma}) + \text{trace}(\boldsymbol{\mu}'\boldsymbol{\mu}) = \text{trace}(\mathbf{\Sigma}) + \boldsymbol{\mu}'\boldsymbol{\mu} \end{aligned}$$

using the facts that the trace is a linear operator, and that if $\mathbf{AB}$ and $\mathbf{BA}$ are both square matrices, then $\text{trace}(\mathbf{AB}) = \text{trace}(\mathbf{BA})$. □

Proposition 6 (Quadratic form)

Let $w = \mathbf{y}'\mathbf{T}\mathbf{y}$, where the $n \times n$ matrix $\mathbf{T}$ is non-random. Then the random variable w has expectation $E(w) = \text{trace}(\mathbf{T}\mathbf{\Sigma}) + \boldsymbol{\mu}'\mathbf{T}\boldsymbol{\mu}$.

Proof.

Write $\mathbf{y}'\mathbf{T}\mathbf{y} = \text{trace}(\mathbf{y}'\mathbf{T}\mathbf{y}) = \text{trace}(\mathbf{T}\mathbf{y}\mathbf{y}') = \text{trace}(\mathbf{T}\mathbf{W})$. Then

$$\begin{aligned} E(\mathbf{y}'\mathbf{T}\mathbf{y}) &= E[\text{trace}(\mathbf{T}\mathbf{W})] = \text{trace}[E(\mathbf{T}\mathbf{W})] = \text{trace}[\mathbf{T}E(\mathbf{W})] \\ &= \text{trace}[\mathbf{T}(\mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}')] = \text{trace}(\mathbf{T}\mathbf{\Sigma}) + \text{trace}(\mathbf{T}\boldsymbol{\mu}\boldsymbol{\mu}') \\ &= \text{trace}(\mathbf{T}\mathbf{\Sigma}) + \boldsymbol{\mu}'\mathbf{T}\boldsymbol{\mu}. \end{aligned}$$



Proposition 7 (Pair of vector linear functions)

Let $\mathbf{z}_1 = \mathbf{g}_1 + \mathbf{H}_1\mathbf{y}$, $\mathbf{z}_2 = \mathbf{g}_2 + \mathbf{H}_2\mathbf{y}$, where $m_1 \times 1$ vector $\mathbf{g}_1$, the $m_2 \times 1$ vector $\mathbf{g}_2$, the $m_1 \times n$ matrix $\mathbf{H}_1$, and the $m_2 \times n$ matrix $\mathbf{H}_2$ are non-random. Then $C(\mathbf{z}_1, \mathbf{z}_2) = \mathbf{H}_1 \Sigma \mathbf{H}_2'$.

Proof.

Let $\mathbf{z}_1^* = \mathbf{z}_1 - E(\mathbf{z}_1) = \mathbf{H}_1\boldsymbol{\epsilon}$, and $\mathbf{z}_2^* = \mathbf{z}_2 - E(\mathbf{z}_2) = \mathbf{H}_2\boldsymbol{\epsilon}$. Then $\mathbf{z}_1^* \mathbf{z}_2^{*'} = \mathbf{H}_1 \boldsymbol{\epsilon} \boldsymbol{\epsilon}' \mathbf{H}_2'$, so

$$C(\mathbf{z}_1, \mathbf{z}_2) = E(\mathbf{z}_1^* \mathbf{z}_2^{*'}) = \mathbf{H}_1 E(\boldsymbol{\epsilon} \boldsymbol{\epsilon}') \mathbf{H}_2' = \mathbf{H}_1 \Sigma \mathbf{H}_2'$$



The assumptions of the linear model in matrix form.

MLR.1 Linear in parameters

The model can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

where $\mathbf{y}$ is a $n \times 1$ vector collecting the observations of the dependent variable, $\mathbf{X}$ is an $n \times (k+1)$ matrix of observed explanatory variables (which includes a constant term) and $\mathbf{u}$ is a vector of unobserved disturbances.

MLR.2 Random Sampling

The observations in $\mathbf{y}$ and $\mathbf{X}$ come from a random sample of size n from the population.

MLR.3 $\text{rank}(\mathbf{X}) = k + 1$, i.e. no perfect collinearity

The matrix of $\mathbf{X}$ has rank $k + 1$, i.e., there are $k + 1$ linearly independent columns. This ensures that the square $(k + 1) \times (k + 1)$ matrix $\mathbf{Q} = \mathbf{X}'\mathbf{X}$ also has rank $k + 1$ and thus is invertible.

MLR.4 Zero Conditional Mean

Conditional on the entire matrix $\mathbf{X}$, each error $E(u_i|\mathbf{X}) = 0$ for $i = 1, \dots, n$. Written in vector form

$$E(\mathbf{u}|\mathbf{X}) = \mathbf{0}.$$

This implies, using MLR.1, MLR.2 and MLR.3, that

$$E(\mathbf{y}|\mathbf{X}) = \mathbf{X}\boldsymbol{\beta},$$

and using the law of iterated expectations, that $E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$.

Unbiasedness of $\hat{\beta}$

- Recall the expressions for the following matrices:

$$\mathbf{A} = \mathbf{Q}^{-1}\mathbf{X}', \quad \mathbf{N} = \mathbf{X}\mathbf{A}, \quad \mathbf{M} = \mathbf{I} - \mathbf{N}.$$

and what they do:

$$\hat{\beta} = \mathbf{A}\mathbf{y}, \quad \hat{\mathbf{y}} = \mathbf{N}\mathbf{y}, \quad \hat{\mathbf{u}} = \mathbf{M}\mathbf{y}.$$

- The matrices $\mathbf{Q}$, $\mathbf{A}$, $\mathbf{N}$ and $\mathbf{M}$ only depend on $\mathbf{X}$.

Proposition 8 (Unbiasedness of the OLS estimator)

Under assumptions MLR1–MLR4 we have that $E(\hat{\beta}) = \beta$.

Proof.

We start by calculating the conditional expectation:

$E(\hat{\beta}|\mathbf{X}) = E(\mathbf{A}\mathbf{y}|\mathbf{X}) = \mathbf{A}E(\mathbf{y}|\mathbf{X}) = \mathbf{A}\mathbf{X}\beta = \mathbf{I}\beta = \beta$, since $\mathbf{A}\mathbf{X} = \mathbf{I}$. Using the law of iterated expectations, we obtain that $E(\hat{\beta}) = \beta$. □

Variance of the OLS estimator

- To compute the variance of the OLS estimator, we will make the extra assumption of homoskedasticity

MLR.5 Homoskedasticity

This assumption is composed of two parts:

- 1 $V(u_i|\mathbf{X}) = \sigma^2$ for $i = 1, \dots, n$
- 2 $Cov(u_i, u_h) = 0$, for all $i \neq h$.

In matrix form, we can write these two assumptions as

$$V(\mathbf{u}|X) = \sigma^2 \mathbf{I}_n,$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix. Using proposition 3 (vector linear function), MLR4 implies that

$$V(\mathbf{y}|X) = \sigma^2 \mathbf{I}_n.$$

Proposition 9 (Variance-Covariance matrix of the OLS estimator)

Under assumptions MLR1–MLR5, we have that

$$V(\hat{\beta}|X) = \sigma^2 \mathbf{Q}^{-1} = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$$

^a *The variances and covariances of the $k+1$ random variables in $\hat{\beta}$ are given by the appropriate elements of the $(k+1) \times (k+1)$ matrix $\sigma^2 \mathbf{Q}^{-1}$:*

$$V(\hat{\beta}_j|\mathbf{X}) = \sigma^2 q^{jj}, \quad C(\hat{\beta}_h, \hat{\beta}_j|\mathbf{X}) = \sigma^2 q^{hj},$$

where q^{hj} denotes the element in the h th row and j th column of $\mathbf{Q}^{-1}$.

^aYou can check the other set of slides (Chapter3-part2-EC0301-Spring2020.pdf, slide 55) for a version of this theorem without matrix notation. I believe the version I give you here is much more elegant and more useful for future calculations.

Proof.

Using proposition 2: $V(\hat{\beta}|X) = V(\mathbf{A}\mathbf{y}|\mathbf{X}) = \mathbf{A} V(\mathbf{y}|\mathbf{X}) \mathbf{A}' = \mathbf{A} \sigma^2 \mathbf{I}_n \mathbf{A}' = \sigma^2 \mathbf{A} \mathbf{A}' = \sigma^2 \mathbf{Q}^{-1}$, where $\mathbf{A} \mathbf{A}' = \mathbf{Q}^{-1}$ (by property 4.) □

Estimation of σ^2 and $V(\hat{\beta}|\mathbf{X})$

- In the expression for the variance, σ^2 is also an unknown parameter, so we will need to estimate it from the data.
- The estimation of σ^2 is done by drawing on the least squares residual vector $\hat{\mathbf{u}} = \mathbf{M}\mathbf{y}$.
- $\mathbf{M}$ is a function of $\mathbf{X}$ alone.
- Thus, $\hat{\mathbf{u}} = \mathbf{M}\mathbf{y}$ is a linear function of the vector $\mathbf{y}$ so proposition 2 applies.
- Remark: All expectations in this part are conditional expectations with respect to $\mathbf{X}$. We have that
 - ▶ $E(\hat{\mathbf{u}}) = \mathbf{M}E(\mathbf{y}) = \mathbf{M}\mathbf{X}\beta = (\mathbf{M}\mathbf{X})\beta = \mathbf{O}\beta = \mathbf{0}$, since $\mathbf{M}\mathbf{X} = \mathbf{O}$.
 - ▶ $V(\hat{\mathbf{u}}) = \mathbf{M}V(\mathbf{y})\mathbf{M}' = \mathbf{M}(\sigma^2\mathbf{I}_n)\mathbf{M}' = \sigma^2\mathbf{M}\mathbf{M}' = \sigma^2\mathbf{M}$, since $\mathbf{M}$ is idempotent.
- The estimator of σ^2 will be

$$\hat{\sigma}^2 = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n - k - 1}$$

Proposition 10 (Unbiasedness of $\hat{\sigma}^2$)

Under assumptions MLR1–MLR5, $\hat{\sigma}^2$ is unbiased for σ^2 : $E(\hat{\sigma}^2) = \sigma^2$ for all $\sigma^2 > 0$.

Proof.

I start by computing the expectation of the random variable $\hat{\mathbf{u}}'\hat{\mathbf{u}}$, which is the sum of squared residuals. Apply proposition 5 with $\hat{\mathbf{u}}$ playing the role of $\mathbf{y}$ in the statement of the result:

$$E(\hat{\mathbf{u}}'\hat{\mathbf{u}}) = \text{trace}(V(\hat{\mathbf{u}})) + [E(\hat{\mathbf{u}})]'[E(\hat{\mathbf{u}})] = \text{trace}(\sigma^2\mathbf{M}) + \mathbf{0}'\mathbf{0} = \sigma^2 \text{trace}(\mathbf{M})$$

But $\mathbf{N} = \mathbf{X}\mathbf{A}$ and $\mathbf{A}\mathbf{X} = \mathbf{I}_{k+1}$, so

$$\text{trace}(\mathbf{N}) = \text{trace}(\mathbf{X}\mathbf{A}) = \text{trace}(\mathbf{A}\mathbf{X}) = k + 1.$$

Hence, for $\mathbf{M} = \mathbf{I}_n - \mathbf{N}$ we have that

$$\text{trace}(\mathbf{M}) = \text{trace}(\mathbf{I}_n - \mathbf{N}) = \text{trace}(\mathbf{I}_n) - \text{trace}(\mathbf{N}) = n - k - 1.$$

Thus $E(\hat{\mathbf{u}}'\hat{\mathbf{u}}) = \sigma^2(n - k - 1)$. Therefore, $E(\hat{\sigma}^2) = E(\hat{\mathbf{u}}'\hat{\mathbf{u}})/(n - k - 1) = \sigma^2$. □

- Thus, we will estimate the variance matrix $V(\hat{\beta}|\mathbf{X}) = \sigma^2\mathbf{Q}^{-1}$ by

$$\widehat{V(\hat{\beta})} = \hat{\sigma}^2\mathbf{Q}^{-1}.$$

- Since $\hat{\sigma}^2$ is unbiased and conditional on $\mathbf{X}$, $\mathbf{Q}^{-1}$ is a constant, we have that

$$E\left(\widehat{V(\hat{\beta})}\right) = \sigma^2\mathbf{Q}^{-1} = V(\hat{\beta}),$$

so that $\widehat{V(\hat{\beta})}$ is an unbiased estimator of $V(\hat{\beta})$.

- In particular $\hat{\sigma}^2 q^{jj}$ is an unbiased estimator of $V(\hat{\beta}_j) = \sigma_{\beta_j}^2 = \sigma^2 q^{jj}$.
- The square root of the estimated variance $\hat{\sigma}_{\beta_j} = \hat{\sigma}\sqrt{q^{jj}}$ is the **standard error of β_j**

Example with WAGE2 data

```
> model4<-lm(lwage~educ+ IQ + exper,data=wage2)
> summary(model4)
```

```
Call:
lm(formula = lwage ~ educ + IQ + exper, data = wage2)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.93902 -0.21414  0.02295  0.26270  1.25771
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.1980851  0.1215426  42.768  < 2e-16 ***
educ         0.0571080  0.0073480   7.772 2.04e-14 ***
IQ           0.0057856  0.0009797   5.906 4.92e-09 ***
exper        0.0195249  0.0032444   6.018 2.53e-09 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.3861 on 931 degrees of freedom
Multiple R-squared:  0.1622, Adjusted R-squared:  0.1595
F-statistic: 60.1 on 3 and 931 DF, p-value: < 2.2e-16
```

- After running the model, we can extract the estimated variance covariance matrix $\widehat{V(\hat{\beta})}$ with the following commands.

```
> Vhat<-vcov(model)
> sqrt(Vhat[1,1])
[1] 0.1215426
> Vhat
```

	(Intercept)	educ	IQ	exper
(Intercept)	1.477261e-02	-4.919911e-04	-5.053159e-05	-2.480936e-04
educ	-4.919911e-04	5.399248e-05	-3.429426e-06	9.697334e-06
IQ	-5.053159e-05	-3.429426e-06	9.597302e-07	-4.177751e-08
exper	-2.480936e-04	9.697334e-06	-4.177751e-08	1.052583e-05

- Focus on the element (2, 2) of the matrix above, it is the estimated variance of the coefficient on educ. If we take the square root, we will get the standard error that is reported in the usual summary of the estimation.

```
> sqrt(Vhat[2,2])
[1] 0.007347957
```

- The diagonal elements are the estimated variances of the coefficients and the off-diagonal elements are the estimated covariances.

The Gauss-Markov Theorem

- We already know that under assumptions MLR1–MLR4, the OLS estimator is unbiased.
- However, why do we prefer this estimator? Why not use another one?
- It turns out that under assumptions MLR1–MLR4 plus MLR5 (homoskedasticity), the OLS estimator is the **minimum variance linear unbiased estimator**, or the *best linear unbiased estimator (BLUE)*.

Theorem 11 (The Gauss-Markov Theorem)

Under assumptions MLR1–MLR5, $\hat{\beta}$ is the best linear unbiased estimator of the parameter vector β .

Proof.

Let $\mathbf{b}^* = \mathbf{A}^* \mathbf{y}$, where $\mathbf{A}^*$ is any $(k+1) \times n$ nonstochastic matrix. Then $\mathbf{b}^*$ is a linear function of $\mathbf{y}$, so this is another linear estimator. Under proposition 2, we have

$$\begin{aligned} E(\mathbf{b}^*) &= \mathbf{A}^* E(\mathbf{y}) = \mathbf{A}^* \mathbf{X} \boldsymbol{\beta}, \\ V(\mathbf{b}^*) &= \mathbf{A}^* V(\mathbf{y}) \mathbf{A}^{*'} = \sigma^2 \mathbf{A}^* \mathbf{A}^{*'} \end{aligned}$$

Clearly, $\mathbf{b}^*$ will be unbiased for $\boldsymbol{\beta}$ if and only if $\mathbf{A}^* \mathbf{X} = \mathbf{I}$. Write $\mathbf{A}^* = \mathbf{A} + \mathbf{D}$, where $\mathbf{A} = \mathbf{Q}^{-1} \mathbf{X}'$ and $\mathbf{D} = \mathbf{A}^* - \mathbf{A}$. Notice that

$$\begin{aligned} \mathbf{A}^* \mathbf{X} &= \mathbf{A} \mathbf{X} + \mathbf{D} \mathbf{X} = \mathbf{I} + \mathbf{D} \mathbf{X} \\ \mathbf{A}^* \mathbf{A}^{*'} &= (\mathbf{A} + \mathbf{D})(\mathbf{A} + \mathbf{D})' = \mathbf{A} \mathbf{A}' + \mathbf{A} \mathbf{D}' + \mathbf{D} \mathbf{A}' + \mathbf{D} \mathbf{D}' \end{aligned}$$

So the unbiasedness condition $\mathbf{A}^* \mathbf{X} = \mathbf{I}$ is equivalent to $\mathbf{D} \mathbf{X} = \mathbf{O}$, which implies that $\mathbf{D} \mathbf{X} \mathbf{Q}^{-1} = \mathbf{O}$, that is $\mathbf{D} \mathbf{A}' = \mathbf{O}$ and $\mathbf{A} \mathbf{D}' = \mathbf{O}$. so if $\mathbf{b}^*$ is an unbiased estimator of $\boldsymbol{\beta}$, then

$$V(\mathbf{b}^*) = \sigma^2 (\mathbf{A} \mathbf{A}' + \mathbf{D} \mathbf{D}') = V(\hat{\boldsymbol{\beta}}) + \sigma^2 \mathbf{D} \mathbf{D}'.$$

The matrix $\mathbf{D} \mathbf{D}'$ is nonnegative definite, and the scalar σ^2 is positive, so $\sigma^2 \mathbf{D} \mathbf{D}'$ is nonnegative definite. Consequently $V(\mathbf{b}^*) \geq V(\hat{\boldsymbol{\beta}})$, with equality if and only if $\mathbf{D} \mathbf{D}' = \mathbf{O}$, that is, if and only if $\mathbf{D} = \mathbf{O}$, that is, if and only if $\mathbf{A}^* = \mathbf{A}$, that is, if and only if $\mathbf{b}^* = \hat{\boldsymbol{\beta}}$ in every sample. □

Remarks about the proof

- The matrix $\mathbf{D}\mathbf{D}'$ is nonnegative definite because for any $(k+1) \times 1$ vector $\mathbf{h}$, the quadratic form $\mathbf{h}'\mathbf{D}\mathbf{D}'\mathbf{h} = (\mathbf{D}'\mathbf{h})'(\mathbf{D}'\mathbf{h}) = \mathbf{v}'\mathbf{v} \geq 0$.
- If $\mathbf{z}$ and $\mathbf{y}$ are random vectors, we say that $V(\mathbf{z}) \geq V(\mathbf{y})$ if and only if $V(\mathbf{z}) - V(\mathbf{y})$ is nonnegative definite.
- Observe the implications of the nonnegative definiteness of $V(\mathbf{b}^*) - V(\hat{\beta})$. Element by element $\hat{\beta}$ is preferable to $\mathbf{b}^*$, any other linear unbiased estimator of β , because its elements have smaller variances. But also consider a linear combination of the β_j 's, say $\theta = \mathbf{h}'\beta$, where $\mathbf{h}$ is a constant $(k+1) \times 1$ vector. Let $t = \mathbf{h}'\hat{\beta}$ and $t^* = \mathbf{h}'\mathbf{b}^*$. Then both t and t^* are unbiased for θ but $V(t^*) - V(t) = \mathbf{h}'[V(\mathbf{b}^*) - V(\hat{\beta})]\mathbf{h} \geq 0$. So $\hat{\beta}$ is also preferred to $\mathbf{b}^*$ for constructing estimators of linear combinations.

Estimation of Linear functions of $\hat{\beta}$

- Suppose that you have a random vector $\mathbf{y}$. In general, such a vector would have $E(\mathbf{y}) = \boldsymbol{\mu}$ and $V(\mathbf{y}) = \boldsymbol{\Sigma}$. For the multiple linear regression model, we have seen that $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ and $V(\mathbf{y}|\mathbf{X}) = \sigma^2\mathbf{I}$.
- This means that $\mu_i = \mathbf{x}_i'\boldsymbol{\beta}$, $i = 1, \dots, n$.
- In this model, we estimate $\boldsymbol{\beta}$ by $\hat{\boldsymbol{\beta}}$, and thus we estimate $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ by $\hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{N}\mathbf{y} = \hat{\mathbf{y}}$.
- $E(\hat{\mathbf{y}}) = \boldsymbol{\mu}$ and $E(\mathbf{y}) = \boldsymbol{\mu}$ so both are unbiased estimators of $\boldsymbol{\mu}$.
- Why do we use $\hat{\mathbf{y}}$ instead of $\mathbf{y}$ to estimate $\boldsymbol{\mu}$?
- The answer is simple: $V(\mathbf{y}|\mathbf{X}) = \sigma^2\mathbf{I}$ and $V(\hat{\mathbf{y}}|\mathbf{X}) = \sigma^2\mathbf{N}$ so that $V(\mathbf{y}|\mathbf{X}) - V(\hat{\mathbf{y}}|\mathbf{X}) = \sigma^2(\mathbf{I} - \mathbf{N}) = \sigma^2\mathbf{M}$.
- The matrix $\mathbf{M}$ is nonnegative definite so $V(\mathbf{y}|\mathbf{X}) \geq V(\hat{\mathbf{y}}|\mathbf{X})$.
- Thus $\hat{\mathbf{y}}$ is **more efficient** than $\mathbf{y}$.

- For a single element of $\boldsymbol{\mu}$, say μ_i , the preferred estimator is $\hat{\mu}_i = \mathbf{x}_i' \hat{\boldsymbol{\beta}} = \hat{y}_i = \mathbf{n}_i' \mathbf{y}$, where $\mathbf{n}_i'$ denotes the i th row of $\mathbf{N}$.
- A simpler unbiased estimator is $y_i = \mathbf{h}_i' \mathbf{y}$, where $\mathbf{h}_i$ is the $n \times 1$ vector with a 1 in the i th slot and zeroes everywhere else.
- Observe that $\hat{\mu}_i$ is a linear function of all the y 's whereas y_i is trivially a function of only one of them.
- Formally, the reason $\hat{\mu}_i$ is preferred is because of its lower variance.
- Intuitively, $\hat{\mu}_i$, **leverages all of the observations** to produce the estimate, whereas y_i uses only one observation.

- $\mu_i = \mathbf{x}'\boldsymbol{\beta}$ is a special case of a linear combination of the β_j 's. Consider the general case $\theta = \mathbf{h}'\boldsymbol{\beta}$ where $\mathbf{h}$ is a $(k+1) \times 1$ nonrandom vector.
- Examples: $\mathbf{h} = (0, 0, 1, 0, \dots, 0)'$ would give $\theta = \beta_3$; $\mathbf{h} = (0, 1, -1, 0, \dots, 0)'$, then $\theta = \beta_2 - \beta_3$.
- The appropriate estimator for θ is $t = \mathbf{h}'\hat{\boldsymbol{\beta}}$.
- We have that
 - ▶ $E(t) = \mathbf{h}'E(\hat{\boldsymbol{\beta}}) = \mathbf{h}'\boldsymbol{\beta} = \theta$ so t is unbiased for θ .
 - ▶ $V(t) = \mathbf{h}'V(\hat{\boldsymbol{\beta}})\mathbf{h} = \sigma^2\mathbf{h}'\mathbf{Q}^{-1}\mathbf{h}$.
 - ▶ We can express t as a linear function of $\mathbf{y}$: $t = \mathbf{h}'\hat{\boldsymbol{\beta}} = \mathbf{h}'\mathbf{A}\mathbf{y} = \mathbf{w}'\mathbf{y}$ where $\mathbf{w} = \mathbf{A}'\mathbf{h}$ is $n \times 1$ and nonrandom.

- Consider all linear functions of $\mathbf{y}$ that could instead have been used to estimate θ : Let $\mathbf{w}^*$ be an arbitrary $n \times 1$ nonrandom vector, and let $t^* = \mathbf{w}^{*'}\mathbf{y}$, then we have
 - ▶ $E(t^*) = \mathbf{w}^{*'}\mathbf{X}\boldsymbol{\beta}$
 - ▶ $V(t^*) = \sigma^2\mathbf{w}^{*'}\mathbf{w}^*$
- Thus, t^* is unbiased if $\mathbf{w}^{*'}\mathbf{X} = \mathbf{h}'$.
- In this case, we can write $\mathbf{h}'\mathbf{Q}^{-1}\mathbf{h} = \mathbf{w}^{*'}\mathbf{X}\mathbf{Q}^{-1}\mathbf{X}'\mathbf{w}^* = \mathbf{w}^{*'}\mathbf{N}\mathbf{w}^*$
- The previous implies that $V(t) = \sigma^2\mathbf{w}^{*'}\mathbf{N}\mathbf{w}^*$.
- All of the above implies that $V(t^*) - V(t) = \sigma^2\mathbf{w}^{*'}(\mathbf{I} - \mathbf{N})\mathbf{w}^* = \sigma^2\mathbf{w}^{*'}\mathbf{M}\mathbf{w}^* \geq 0$
- Thus, $t = \mathbf{h}'\boldsymbol{\beta}$ as an estimator of θ is a BLUE estimator. *In practice, to estimate $V(t)$, replace σ^2 with $\hat{\sigma}^2$. The resulting standard error for t is*

$$se(t) = \hat{\sigma}_t = \hat{\sigma}\sqrt{\mathbf{h}'\mathbf{Q}^{-1}\mathbf{h}}$$

Estimation of the Conditional expectation, and prediction

- We have seen that to estimate the parameter $\mu_i = E(y_i) = \mathbf{x}_i'\boldsymbol{\beta}$, the estimator of choice was $\hat{y}_i = \mathbf{x}_i'\hat{\boldsymbol{\beta}}$. The expectation and variance of this estimator are $E(\hat{y}_i) = \mu_i$ and $V(\hat{y}_i) = \sigma^2 \mathbf{x}_i' \mathbf{Q}^{-1} \mathbf{x}_i = \sigma^2 n_{ii}$, where n_{ii} is the i th diagonal element of $\mathbf{N}$.
- Suppose that we are interested in estimating $E(y|\mathbf{x}_0) = \mathbf{x}_0'\boldsymbol{\beta} = \mu_0$ where $\mathbf{x}_0$ is some $(k+1) \times 1$ vector, not necessarily part of the original data. Because μ_0 is a linear combination of the elements of $\boldsymbol{\beta}$, as we have seen, the preferred estimator is $\hat{\mu}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$, which has expectation and variance

$$E(\hat{\mu}_0) = \mu_0, \quad V(\hat{\mu}_0) = \sigma^2 \mathbf{x}_0' \mathbf{Q}^{-1} \mathbf{x}_0$$

The standard error for this estimator is $\hat{\sigma} \sqrt{\mathbf{x}_0' \mathbf{Q}^{-1} \mathbf{x}_0}$.

Prediction or forecasting

- Prediction, or forecasting, is a distinct problem.
- Here the objective is to predict the value of y_0 .
- If we knew β , then our prediction would be given by $\mu_0 = x_0' \beta$.
 - ▶ In this case, the prediction error is $\epsilon_0 = y_0 - \mu_0$, with $E(\epsilon_0) = 0$.
 - ▶ The variance would be $V(\epsilon_0) = E(\epsilon_0^2) = V(y_0) = \sigma^2$.
- Since we do not know β but estimate it with $\hat{\beta}$, the natural predictor will be $\hat{\mu}_0 = x_0' \hat{\beta}$.
- When this predictor is used, the prediction error is $u = y_0 - \hat{\mu}_0$, with
 - ▶ $E(u) = E(y_0) - E(\hat{\mu}_0) = 0$.
 - ▶ $V(u) = V(y_0) + V(\hat{\mu}_0) - 2C(y_0, \hat{\mu}_0) = \sigma^2 + \sigma^2 x_0' Q^{-1} x_0 = \sigma^2(1 + x_0' Q^{-1} x_0)$, where the covariance is taken to be zero because the drawing of y_0 is independent of the sample observations in y .
- The variance of the prediction error has two components
 - 1 the variance of the prediction error that would be made were μ_0 be known and used,
 - 2 and the variance of the estimator of μ_0
- The **standard error of forecast** is the square root of the estimate of $V(u)$, i.e.
$$\hat{\sigma} \sqrt{1 + x_0' Q^{-1} x_0}$$

Revisiting the R squared

- Given $\mathbf{y}$ and $\mathbf{X}$, we run the regression to obtain $\hat{\beta}$, $\hat{\mathbf{y}}$ and $\hat{\mathbf{u}}$.
- Observe that $\mathbf{y} = \hat{\mathbf{y}} + \hat{\mathbf{u}}$ and by the FOCs $\hat{\mathbf{y}}'\hat{\mathbf{u}} = (\mathbf{X}\hat{\beta})'\hat{\mathbf{u}} = \hat{\beta}'(\mathbf{X}'\hat{\mathbf{u}}) = \hat{\beta}'\mathbf{0} = \mathbf{0}$, so

$$\mathbf{y}'\mathbf{y} = (\hat{\mathbf{y}} + \hat{\mathbf{u}})'(\hat{\mathbf{y}} + \hat{\mathbf{u}}) = \hat{\mathbf{y}}'\hat{\mathbf{y}} + \hat{\mathbf{u}}'\hat{\mathbf{u}}$$

which is just $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{u}_i^2$. This is just a variance decomposition formula

- Further, $\sum_i y_i = \sum_i \hat{y}_i + \sum_i \hat{u}_i$ so that $\bar{y} = \bar{\hat{y}} + \bar{e}$.
- But since $\bar{e} = 0$, then $\bar{y} = \bar{\hat{y}}$ so $n\bar{y} = n\bar{\hat{y}}$, which if we subtract from the previous variance decomposition formula, we obtain the analysis of variation identity ($TSS = SSE + SSR$):

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{u}_i^2$$

- The R^2 is obtained by dividing the previous equation by $\sum_i (y_i - \bar{y})^2$

$$R^2 = \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2} = 1 - \frac{\sum_i \hat{u}_i^2}{\sum_i (y_i - \bar{y})^2}$$

- One should not calculate the R^2 when the matrix $\mathbf{X}$ does not include the vector of constants as a regressor, i.e. the intercept. The fact that $\bar{e} = 0$ is precisely because the FOCs guarantee this due to the presence of the intercept.

The Adjusted R squared.

- We have seen the interpretation of the R^2 before as a measure of goodness of fit.
- However, as the R^2 tends to increase as additional explanatory variables are included, leads to define an adjusted measure of goodness of fit, the **adjusted R^2** or $\bar{R}^2$ which is defined via

$$1 - \bar{R}^2 = \frac{n-1}{n-k-1}(1 - R^2)$$

- This new measure inflates the unexplained proportion and hence deflates the explained proportion. To see this, here is a more explicit formula

$$\bar{R}^2 = 1 - \frac{\frac{\sum_i \hat{u}_i^2}{n-k-1}}{\frac{\sum_i (y_i - \bar{y})^2}{n-1}}$$

- It is sometimes said that the numerator is an unbiased estimator of the disturbance variance, and that the denominator is an unbiased estimator of the variance of y . The first claim is correct as we now. But the second claim is not correct: under the assumptions we have made about the multiple linear regression model, the variance of the disturbance is the same thing as the common variance of the y , namely, σ^2 .

Short and Long Regression Algebra.

- Given the $n \times 1$ vector $\mathbf{y}$ and the $n \times (k+1)$ matrix $\mathbf{X}$ whose rank is $k+1$, we regress $\mathbf{y}$ on $\mathbf{X}$.
- This produces the coefficients and residual vectors

$$\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}, \quad \hat{\mathbf{u}} = \mathbf{M}\mathbf{y}$$

where $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, $\mathbf{M} = \mathbf{I} - \mathbf{X}\mathbf{A}$ and $\mathbf{A}\mathbf{X} = \mathbf{U}$, $\mathbf{M}\mathbf{X} = \mathbf{O}$, $\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$.

- Partition the set of k_1 explanatory variables in two subsets, $\mathcal{K}_1$ and $\mathcal{K}_2$, each with size k_1 and k_2 respectively, such that $k_1 + k_2 = k + 1$
- Correspondingly, partition $\mathbf{X}$ as $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$, where $\mathbf{X}_1$ is $n \times k_1$ and $\mathbf{X}_2$ is $n \times k_2$.
- Correspondingly, partition $\hat{\boldsymbol{\beta}}$ as $\hat{\boldsymbol{\beta}}_1$ and $\hat{\boldsymbol{\beta}}_2$, each with dimensions $k_1 \times 1$ and $k_2 \times 1$ respectively.
- We can write

$$\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}} = (\mathbf{X}_1, \mathbf{X}_2) \begin{pmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{pmatrix} + \hat{\mathbf{u}} = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2\hat{\boldsymbol{\beta}}_2 + \hat{\mathbf{u}}. \quad (11)$$

- Since $\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$, we know that $\mathbf{X}_1'\hat{\mathbf{u}} = \mathbf{0}$ and $\mathbf{X}_2'\hat{\mathbf{u}} = \mathbf{0}$.

- Suppose we shorten the list of explanatory variables and regress y only on the first k_1 explanatory variables, i.e., on $\mathbf{X}_1$, that is, we must choose $\mathbf{c}_1$ to minimize $\mathbf{u}^{*'}\mathbf{u}$ where $\mathbf{u}^* = \mathbf{y} - \mathbf{X}_1\mathbf{c}_1$
- the resulting coefficient vector and residual vector are

$$\mathbf{b}_1^* = \mathbf{A}_1\mathbf{y} \quad \mathbf{e}^* = \mathbf{M}_1\mathbf{y}$$

where $\mathbf{A} - 1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'$ and $\mathbf{M}_1 = \mathbf{I} - \mathbf{X}_1\mathbf{A}_1$.

- We also have that $\mathbf{A}_1\mathbf{X}_1 = \mathbf{I}$, $\mathbf{M}_1\mathbf{X}_1 = \mathbf{O}$ and $\mathbf{X}_1'\mathbf{e}^* = \mathbf{0}$.
- As a result of running this short regression we have that

$$\mathbf{y} = \mathbf{X}_1\mathbf{b}_1^* + \mathbf{e}^*. \quad (12)$$

- Now we are going to derive the relationship between $\mathbf{b}_1^*$ and $\mathbf{e}^*$ and $\hat{\beta}_1$ and $\hat{\mathbf{u}}$, i.e. we are going to see how the **short regression** (12) is related to the **long regression** (11).

Auxiliary Regressions.

- To obtain the relationship between the short and long regressions, we will construct some **auxiliary regressions**.
- Regress each column of $\mathbf{X}_2$ in turn upon $\mathbf{X}_1$ to obtain a set of auxiliary regressions:

$$\mathbf{x}_j = \mathbf{X}_1 \mathbf{f}_j + \mathbf{x}_j^*, \quad j \in \mathcal{K}_2$$

where $\mathbf{f}_j = \mathbf{A}_1 \mathbf{x}_j$ and $\mathbf{x}_j^* = \mathbf{M}_1 \mathbf{x}_j$.

- Assemble all of the previous regressions as

$$\mathbf{X}_2 = \mathbf{X}_1 \mathbf{F} + \mathbf{X}_2^* \tag{13}$$

where $\mathbf{F} = (\mathbf{f}_j)_{j \in \mathcal{K}_2} = \mathbf{A}_1 \mathbf{X}_2$ is a $k_1 \times k_2$ matrix: each column contains the coefficients from the regression of a column of $\mathbf{X}_2$ upon (all the columns of) $\mathbf{X}_1$ and $\mathbf{X}_2^* = (\mathbf{x}_j^*)_{j \in \mathcal{K}_2} = \mathbf{M}_1 \mathbf{X}_2$ is an $n \times k_2$ matrix, each column of which contains the residuals from the regression of a column of $\mathbf{X}_2$ upon (all the columns of) $\mathbf{X}_1$.

- Because $\mathbf{X}$ has full column rank, so does $\mathbf{X}_2^*$.

- Using equation (11) to calculate

$$\mathbf{b}_1^* = \mathbf{A}_1 \mathbf{y} = \mathbf{A}_1 (\mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + \hat{\mathbf{u}}) = \hat{\boldsymbol{\beta}}_1 + \mathbf{F} \hat{\boldsymbol{\beta}}_2 \quad (14)$$

because $\mathbf{A}_1 \hat{\mathbf{u}} = (\mathbf{X}_1' \mathbf{X}_1)^{-1} \mathbf{X}_1' \hat{\mathbf{u}}$ and $\mathbf{X}_1' \hat{\mathbf{u}} = 0$.

- (14) says that the coefficients of $\mathbf{X}_1$ in the short regression are a mixture of the coefficients of $\mathbf{X}_1$ and on $\mathbf{X}_2$ in the long regression, with the auxiliary regression coefficients serving as weights in the mixture.
- Use again (11) to calculate

$$\mathbf{e}^* = \mathbf{M}_1 \mathbf{y} = \mathbf{M}_1 (\mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + \hat{\mathbf{u}}) = \mathbf{X}_2^* \hat{\boldsymbol{\beta}}_2 + \hat{\mathbf{u}} \quad (15)$$

because $\mathbf{M}_1 \hat{\mathbf{u}} = (\mathbf{I} - \mathbf{X}_1 \mathbf{A}_1) \hat{\mathbf{u}}$ and $\mathbf{A}_1 \hat{\mathbf{u}} = 0$.

- (14) says is that the residuals from the short regression equal the residuals from the long regression plus a mixture of the elements of $\mathbf{X}_2^*$ with the elements of $\hat{\boldsymbol{\beta}}_2$ serving as weights in the mixture.

- We can use (15) along with the facts that $\mathbf{X}_2^* = \mathbf{X}_2' \mathbf{M}_1$, $\mathbf{M}_1 \hat{\mathbf{u}} = \hat{\mathbf{u}}$, and $\mathbf{X}_2' \hat{\mathbf{u}} = \mathbf{0}$ to calculate

$$\mathbf{e}^{*'} \mathbf{e}^* = \hat{\mathbf{u}}' \hat{\mathbf{u}} + \hat{\boldsymbol{\beta}}_2' \mathbf{X}_2^{*'} \mathbf{X}_2^* \hat{\boldsymbol{\beta}}_2.$$

- (42) says that the sum of squared residuals from the short regression exceeds the sum of squared residuals from the long regression by the nonnegative quantity $\mathbf{v}'\mathbf{v}$ where $\mathbf{v} = \mathbf{X}_2^* \hat{\boldsymbol{\beta}}_2$.
- This result is what is behind the fact that shortening a regression cannot improve the fit of the model and the reason why, adding more explanatory variables increases the R^2 of the model.
- The two sums of squared residuals are equal if and only if $\mathbf{v} = \mathbf{0}$, that is, if and only if $\mathbf{X}_2^* \hat{\boldsymbol{\beta}}_2 = \mathbf{0}$, and since $\text{rank}(\mathbf{X}_2^*) = k_2$, if and only if $\hat{\boldsymbol{\beta}}_2 = \mathbf{0}$.
- Building up on the previous, we can see that the short and long regressions deliver different estimates, but we have two exceptional cases
 - 1 If $\hat{\boldsymbol{\beta}}_2 = \mathbf{0}$, then $\mathbf{b}_1^* = \hat{\boldsymbol{\beta}}_1$, $\mathbf{e}^* = \hat{\mathbf{u}}$ and $\mathbf{e}^{*'} \mathbf{e}^* = \hat{\mathbf{u}}' \hat{\mathbf{u}}$.
 - 2 If $\mathbf{X}_1' \mathbf{X}_2 = \mathbf{O}$, then $\mathbf{F} = \mathbf{A}_1 \mathbf{X}_2 = \mathbf{O}$ and $\mathbf{X}_2^* = \mathbf{X}_2$, so $\mathbf{b}_1^* = \hat{\boldsymbol{\beta}}_1$, even though $\mathbf{e}^* \neq \hat{\mathbf{u}}$. This case corresponds to the situation in which every explanatory variable in $\mathbf{X}_1$ is **orthogonal** to every variable in $\mathbf{X}_2$.

Residual Regression and the Frisch-Waugh Theorem.

- We usually talk about multiple linear regression using the *partialling out interpretation*: when we run the long regression of y on X_1 and X_2 rather than a short regression on X_2 alone, to get coefficients on X_2 , we say that we are getting the effect of X_2 **after controlling for X_1** or after “partialling out X_1 ” or “after holding X_1 constant. ”
- In this section, I develop some algebra to give content to that language.
- Consider the **residual regression**:
 - ▶ Regress y on $X_2^* = M_1 X_2$, the $n \times k_2$ matrix of residuals from the auxiliary regressions of X_2 on X_1 .
 - ▶ The coefficient vector will be $c_2 = A_2^* y$, where $A_2^* = (X_2^{*'} X_2^*)^{-1} X_2^{*'}$ or $(X_2^{*'} X_2^*)^{-1} X_2' M_1$.
 - ▶ We have that

$$\begin{aligned} M_1 y &= M_1 (X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + \hat{u}) = X_2^* \hat{\beta}_2 + \hat{u}, \\ X_2^{*'} X_2^* &= X_2' M_1 X_2 = X_2 M_1' M_1 X_2 = X_2^{*'} X_2^*, \\ X_2^{*'} \hat{u} &= 0 \end{aligned}$$

- ▶ Thus we have

$$c_2 = A_2^* y = \hat{\beta}_2$$

- Summarizing, $\hat{\beta}_2$ can be obtained by a two-step procedure:
 - 1 Regress $\mathbf{X}_2$ on $\mathbf{X}_1$ to get the residuals $\mathbf{X}_2^*$,
 - 2 Regress $\mathbf{y}$ on $\mathbf{X}_2^*$ to get the coefficients $\hat{\beta}_2$
- Because $\mathbf{A}_2^* \mathbf{M}_1 = \mathbf{A}_2^*$ it is also the case that we obtain the same $\hat{\beta}_2$ by a **double residual regression**, regressing $\mathbf{y}^* = \mathbf{M}_1 \mathbf{y}$ on $\mathbf{X}_2^*$.
- In any case, with $\hat{\beta}_2$ in hand, we can complete the calculation of the long-regression coefficient vector $\hat{\beta}$ by regressing $\mathbf{y}$ on $\mathbf{X}_1$ alone, obtaining $\mathbf{b}_1^* = \mathbf{A}_1 \mathbf{y}$, and then recovering $\hat{\beta}_1$ as $\hat{\beta}_1 = \mathbf{b}_1^* - \mathbf{F} \hat{\beta}_2$ where $\mathbf{F} = \mathbf{A}_1 \mathbf{X}_2$.
- All of the above, gives content to the language used above, i.e., the partialling out interpretation of multiple regression. Indeed $\mathbf{c}_2$ relates $\mathbf{y}$ to $\mathbf{X}_2$ after controlling or partialling out the effects of $\mathbf{X}_1$.
- The above results are known collectively as the **Frisch-Waugh theorem**.

Applications of residual regression.

Trend Removal A popular specification for economic time series involves including a linear time trend as part of the explanatory variables. Should the trend term be included in the regression or should the variables first be “detrended” and then used without the terms included? Frisch and Waugh in 1933 concluded that it does not matter. Indeed, this is an application of the theorem that bears their name.

Seasonal adjustment Some macroeconomic variables show a seasonal pattern. It is customary to deseasonalize or seasonally adjust such variables. A conventional way to do this is to calculate the seasonal means and express each observation as a deviation from its seasonal means. Suppose that each year has four quarters or seasons. We can perform this procedure by regressing the data on four dummy variables representing each quarter. The residuals from this regression form the seasonally adjusted series.

Seasonal Data Suppose we are interested in the regression relation between y and some variables X_2 . Ordinarily, we would include an intercept, i.e., an explanatory variable $s = (1, \dots, 1)'$ in addition to X_2 to get an intercept and a set of slopes. However, if y and X_2 have seasonal components in them, we may want to remove them first. We did this for y in the previous point. To deseasonalize X_2 , regress X_2 on the set of season dummy variables X_1 .

Application of Short vs Long Regression: Omitted Variable Bias

- Perhaps one of the best applications of the algebra we just developed is to calculate the **omitted variable bias** of not including an explanatory variable in a multiple linear regression model.
- Recall the coefficient vector from the short regression $\mathbf{b}_1^* = \mathbf{A}_1 \mathbf{y}$. Apply proposition 2 to get

$$E(\mathbf{b}_1^*) = \mathbf{A}_1 E(\mathbf{y}) = \mathbf{A}_1 (\mathbf{X}_1 \beta_1 + \mathbf{X}_2 \beta_2) = \beta_1 + \mathbf{F} \beta_2 \quad (16)$$

$$V(\mathbf{b}_1^*) = \mathbf{A}_1 V(\mathbf{y}) \mathbf{A}_1' = \sigma^2 \mathbf{A}_1 \mathbf{A}_1' = \sigma^2 (\mathbf{X}_1' \mathbf{X}_1)^{-1} \quad (17)$$

- From (16) we conclude that $\mathbf{b}_1^*$ is a biased estimator of β_1 , where the term $\mathbf{F} \beta_2$ is the **omitted variable bias**. There are two exceptional cases:
 - 1 **Irrelevant Omitted Variable:** If $\beta_2 = \mathbf{0}$, then there is no bias and $\mathbf{b}_1^*$ and $\hat{\beta}_1$ do not differ in any way and they are both unbiased.
 - 2 **Orthogonal Explanatory Variables:** If $\mathbf{F} = \mathbf{O}$, then $\mathbf{b}_1^*$ and $\hat{\beta}_1$ coincide in every sample.
- Rewrite equation (12) as $\hat{\beta}_1 = \mathbf{b}_1^* - \mathbf{F} \hat{\beta}_2$. Then $C(\mathbf{b}_1^*, \hat{\beta}_2) = \mathbf{A}_1 V(\mathbf{y}) \mathbf{A}_2^{*'} = \sigma^2 \mathbf{A}_1 \mathbf{A}_2^{*'} = \mathbf{O}$, because $\mathbf{M}_1 \mathbf{X}_2 = \mathbf{O}$ implies $\mathbf{A}_2^* \mathbf{A}_1' = \mathbf{O}$.
- Consequently, $V(\hat{\beta}_1) = V(\mathbf{b}_1^*) + \mathbf{F} V(\hat{\beta}_2) \mathbf{F}'$. Since $\mathbf{F} V(\hat{\beta}_2) \mathbf{F}'$ is nonnegative definite, we have $V(\hat{\beta}_1) \geq V(\mathbf{b}_1^*)$.
- The previous, implies that there is a bias-variance trade-off between the short and long regressions when the target of interest is $\hat{\beta}_1$, i.e., if we are only interested in $\hat{\beta}_1$, the estimate from the short regression may have lower variance than the one from the long regression.